

Umut Onur Yaşar

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Summary

Applied AI Research Engineer specializing in knowledge distillation and parameter-efficient fine-tuning. First-author arXiv preprint on KD methodology; merged contributor to Hugging Face PEFT. Runs end-to-end research — experiment design, controlled ablations, publication — with efficiency under hardware constraints as a research focus rather than a limitation.

Stack: PyTorch · CUDA · Python · Linux · Git

Publications

2026 **Student Capacity Moderates Knowledge Distillation Effectiveness: A Systematic Study Across ResNet Teacher-Student Pairs on CIFAR-10**, *arXiv:2605.31191*

Controlled comparison of Logit-KD and Feature-KD across four ResNet teacher-student pairs (3 seeds, two-stage hyperparameter selection). Student capacity — not teacher-student accuracy gap — emerges as the primary moderator of KD effectiveness; Feature-KD matches or outperforms Logit-KD across all pairs. v2 retracts a v1 bug attribution after a controlled re-run, with the revision history documented.

arxiv.org/abs/2605.31191 · github.com/umutonuryasar/kd-capacity-gap

Open-Source Contributions

2026 **Hugging Face PEFT**, *PR #3293 (merged)*

Fixed a CUDA memory caching bug in `prepare_model_for_kbit_training`; revised through maintainer review and merged into main.

github.com/huggingface/peft/pull/3293

2026 **aisuite (andrewyng)**, *PR #319 (merged)*

Python 3.14 compatibility via `docstring-parser` version constraint relaxation.

github.com/andrewyng/aisuite/pull/319

Projects

2026 **RT-DETR Knowledge Distillation — COCO (ongoing)**

Distillation framework for transformer-based real-time object detection. Implements Hungarian-matching-based query alignment, corrected logit-KD loss formulation, and cross-scale feature alignment; systematic ablation suite on a COCO subset in progress, targeting a technical report with deployment benchmarks (ONNX/TensorRT). Developed end-to-end on a single 4 GB GPU.

PyTorch · COCO · RT-DETR · github.com/umutonuryasar/rt-detr-kd

Experience

2024–2025 **AI Trainer (Freelance)**, *Outlier*, Remote

Evaluated, ranked, and annotated LLM outputs for RLHF training data pipelines; wrote and reviewed prompts and reference responses across technical domains.

2015 **Computer Vision Intern**, *HAVELSAN*, Ankara, Turkey

Developed a flight simulation interface in Python with OpenCV-based computer vision modules; researched ROS integration.

2014 **Avionics Intern**, *TAI (Turkish Aerospace)*, Ankara, Turkey

Built a MATLAB-based image recognition system for helicopter landing zone detection; modeled electrical cable layouts in CATIA.

Education

2020 **B.Sc. Electrical & Electronics Engineering**, *Abant İzzet Baysal University*, Bolu, Turkey

Technical Skills

Frameworks PyTorch, Hugging Face (PEFT, Transformers), OpenCV, NumPy, scikit-learn

Optimization Knowledge distillation, PEFT/LoRA, quantization, pruning, ONNX, TensorRT

Programming Python, C/C++, MATLAB, Bash

Tools & Languages CUDA, Git, Docker, Linux · Turkish (Native), English (Full Professional Proficiency)

Certifications Machine Learning (Stanford / Andrew Ng, Coursera) · IBM AI Engineering Professional Certificate